MA1503 Midterm Tutorial

Date: February 27, 2026

Examination Announcement: A midterm test is scheduled for March 7, 2026. It will cover the first 6 lectures. No calculators, books, notes, or electronic devices will be allowed.

Problems

1. For $c \in \mathbb{R}$, let
$$A = \begin{bmatrix} 1 & -1 & 1 \\ -1 & -2 & c \\ 1 & c & -2 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}.$$
Find all $c \in \mathbb{R}$ such that the linear system $Ax = b$
   (a) has no solution,
   (b) has a unique solution,
   (c) has infinitely many solutions. In this last case, find the solution set.

2. Let A be the matrix given by
$$A = \begin{bmatrix} 1 & 2 & 2 & 3 \\ -1 & -2 & 1 & 3 \\ 2 & 4 & 4 & 6 \end{bmatrix}.$$
   (a) Find the Reduced Row Echelon Form (RREF) of the matrix A.
   (b) Let R be the RREF of A. Then $R = EA$ for some invertible matrix E. Express E as a product of elementary matrices.
   (c) Find the solution set for the equation $Ax = 0$. Express the solution set as the span of two vectors.
   (d) Find a basis of the column space of A.
   (e) Find the rank of A.

3. Find a basis for the minimal subspace in $\mathbb{R}^4$ containing the points $(1, -1, 0, 0)$, $(0, 1, 0, -1)$, $(0, 0, -1, 1)$, $(-1, 0, 1, 0)$.

4. Let u, v be a basis for $\mathbb{R}^2$. Show that
   (a) $u + v, u + v$ is not a basis for $\mathbb{R}^2$;
   (b) $u + v, v$ is a basis for $\mathbb{R}^2$;
   (c) $u + v, u - v$ is a basis for $\mathbb{R}^2$.

5. Suppose v_1, v_2, v_3 are linearly independent vectors.
   (a) Find all scalars a and b such that
$$2av_1 - v_2 = v_1 + bv_2.$$
   (b) Find all scalars k such that the vectors
$$v_1 + 3v_3,\ 2v_1 + kv_3,\ 2v_2$$
   are linearly dependent.

6. Are the following transformations linear?

   (a) $T\begin{bmatrix} x \\ y \end{bmatrix} = x\begin{bmatrix} 1 \\ 2 \end{bmatrix} + 7\begin{bmatrix} x-y \\ 3 \end{bmatrix}$

   (b) $K\begin{bmatrix} x \\ y \end{bmatrix} = x\begin{bmatrix} 1 \\ 2+y \end{bmatrix} + \begin{bmatrix} y \\ x \end{bmatrix}$

   (c) $S\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ |x| \end{bmatrix}$

7. Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Let $\mathcal{B} = \{v_1, v_2, \ldots, v_k\} \subset \mathbb{R}^n$ and $T(\mathcal{B}) = \{T(v_1), \ldots, T(v_k)\}$.

   (a) Assume that T is one-to-one. Show that if $\mathcal{B}$ is linearly independent, then $T(\mathcal{B})$ is linearly independent set of size k.

   (b) Suppose that T is invertible and $\mathcal{B}$ is a basis of $\mathbb{R}^n$. How are k, n, m related? Show that $T(\mathcal{B})$ is a basis of $\mathbb{R}^m$.

8. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ denote the linear transformation which rotates a vector counterclockwise by 45 degrees and then reflects the rotated vector across the y-axis. Find the standard matrix of T.

9. Let A be a 5×5 matrix with the (i, j)-entry $a^i_j \in \mathbb{R}$. Recall that the determinant of A is defined as

$$\det(A) = \sum_{\sigma \in S_5} \operatorname{sgn}(\sigma) a_1^{\sigma(1)} a_2^{\sigma(2)} a_3^{\sigma(3)} a_4^{\sigma(4)} a_5^{\sigma(5)}.$$

   Let $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 1 & 3 & 2 \end{pmatrix} \in S_5$ be a permutation.

   (a) Find all inversions of σ. Use it to compute $\operatorname{sgn}(\sigma)$.

   (b) Write σ as a composition of transpositions. Use it to compute $\operatorname{sgn}(\sigma)$.

10. Consider the permutation $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 3 & 1 & 5 & 2 \end{pmatrix}$.

    (a) Find all inversions of σ. Compute $\operatorname{sgn}(\sigma)$ using the information.

    (b) Write σ as a composition of transpositions. Determine $\operatorname{sgn}(\sigma)$ based on the composition.

    (c) Compute $(1\ 4) \circ \sigma$ and $\sigma \circ (1\ 4)$. Are they the same?

    (d) Compute $\operatorname{sgn}((1\ 4) \circ (2\ 3) \circ (2\ 5))$ and $\operatorname{sgn}(\sigma \circ (1\ 4) \circ (2\ 3) \circ (2\ 5))$.

    (e) Write the transposition $(1\ 5)$ as a composition of adjacent transpositions.

11. Find all possible values of $\operatorname{rank}(A)$ as a varies, where

$$A = \begin{bmatrix} 1 & 2 & a \\ -2 & 4a & 2 \\ a & -2 & 1 \end{bmatrix}.$$